

LPEdit: Locality-Preserving Knowledge Editing for MultiModal Large Language Models

Tianyu Zhang
University of Science and Technology
of China
Hefei, Anhui, China
tianyuz1@mail.ustc.edu.cn

Junfeng Fang*
National University of Singapore
Singapore, Singapore
fangjf@nus.edu.sg

Houcheng Jiang
University of Science and Technology
of China
Hefei, Anhui, China
jianghc@mail.ustc.edu.cn

Xingyu Zhu
University of Science and Technology
of China
Hefei, Anhui, China
xyzhuxyz@mail.ustc.edu.cn

Xiang Wang
University of Science and Technology
of China
Hefei, Anhui, China
xiangwang1223@gmail.com

Xiangnan He
MoE Key Lab of BIPC, University of
Science and Technology of China
Hefei, Anhui, China
xiangnanhe@gmail.com

Abstract

Large Language Models (LLMs) and Vision Large Language Models (VLLMs) demonstrate impressive abilities in comprehending natural language and interpreting visual information, but they can also preserve outdated or incorrect information in both forms. Existing knowledge editing methods can efficiently update erroneous text information in LLMs, avoiding the need for full retraining. The locality in knowledge editing refers to that editing should affect only the targeted outputs while preserving the model's behavior on unrelated inputs. Existing multimodal methods often overlook this principle and do not explicitly design to preserve the consistency of unrelated information in both textual and visual modalities. Here, we propose LPEdit, a novel method that leverages the null space projection on key layers to focus editing on conveyed visual information without influencing unrelated knowledge. Experiments show that our method achieves strong performance across different models and datasets. Moreover, our work advances the understanding and development of locality in multimodal knowledge editing.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Multimedia information systems**.

Keywords

Multimodal Large Language Models, Knowledge Editing, Model Explainability

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1 Introduction

LLMs store vast amounts of factual knowledge acquired from large-scale pretraining corpora [2, 19]. However, these corpora often contain outdated or incorrect information, prompting growing interest in model editing techniques that can efficiently revise specific knowledge without full retraining. This line of work addresses the need for dynamic updates in deployed models while avoiding the high cost of retraining [21]. Recent model editing efforts have mainly focused on the text modality and can be broadly classified into two categories. The first directly modifies model parameters to embed new knowledge [11, 15–17], while the second introduces external mechanisms (e.g., memory modules or adapters) without altering internal weights [18, 24].

Editing VLLMs introduces unique challenges due to the complex interactions between visual and textual modalities, making it a relatively nascent yet rapidly evolving research area. Prior work has extended text-based editing methods to VLLMs [3, 6], but results suggest that such editors are often suboptimal due to cross-modal dependencies beyond decoder weights. While recent methods have made notable progress in ensuring the correctness of edited answers, they often struggle to preserve stable behavior on unrelated samples and tasks. As a result, they may introduce unintended failures, such as off-topic responses, factual interference, or linguistic degradation [9]. These limitations underscore the need to satisfy the accuracy of edited outputs and the reliability of untargeted samples and other tasks at the same time.

To achieve this, we introduce LPEdit, which adopt null space projection to ensure that the model's performance on unrelated inputs remains unaffected while correcting the target outputs. By applying null space projection to the MLP (Multi-Layer Perceptron) projection matrix, we constrain the parameter updates to occur along directions without interfering with prior knowledge. Our method effectively guarantees that the edited model produces correct outputs for target inputs while maintaining stability on unrelated inputs. In this way, our approach not only corrects errors of editing outputs but also avoids introducing biases to irrelevant tasks or degrading the model's essential capabilities. Specifically, in multimodal tasks, we ensure that the interaction between visual and textual information is not disrupted by unnecessary interference.

Our method demonstrates strong performance across two distinct VLLM architectures, highlighting its generality and effectiveness. In particular, we observe consistent improvements in locality, the ability to constrain changes to the target region without affecting unrelated outputs. We show that LPEdit enables effective editing while preserving the integrity of prior knowledge, contributing to the exploration of locality in multimodal knowledge editing.

2 Related Work

Model Editing for LLMs: Model editing for large language models (LLMs) has advanced rapidly, and existing approaches can be broadly divided into categories of approaches that either modify or preserve model parameters. The former directly rewrites internal weights to embed new knowledge, including fine-tuning based instruction-driven modification in LTE [11], hypernetwork-based prediction of targeted weight changes in MEND [17], and causal-tracing-guided interventions such as ROME [15] and its multi-fact extension MEMIT [16]. In contrast, parameter-preserving strategies leverage external mechanisms to steer model behavior without altering weights, such as IKE [24] which performs in-context adjustments, and SERAC [18] which employs a counterfactual model to isolate edits, while T-Patcher [10] adds trainable neurons.

Model Editing for VLLMs: While text-only model editing has achieved notable maturity, extending editing techniques to vision-language large language models (VLLMs) introduces new challenges due to visual-textual grounding and cross-modal consistency. VLLMs typically pair a frozen or lightly tuned language backbone with a pretrained vision encoder, usually a Vision Transformer (ViT) that extracts high-level image features [7]. Architecturally, VLLMs can be grouped into Modal Deep Fusion (MDF) and Modal Early Fusion (MEF) designs [8]: MDF models like ViLBERT [14] inject visual information into intermediate layers via cross-modal attention, while MEF approaches map visual embeddings directly into the language input stream before decoding, as in BLIP-2's Q-Former mechanism [12] and LLaVA's MLP-based alignment [13]. Despite these advances, multimodal model editing remains in its early stage, with MMEdit [6] providing the first benchmark by constructing dedicated editing datasets and evaluation metrics. UniKE [20] introduced a unified memory-based editing framework that disentangles semantic meaning and factual correctness for more reliable updates. VisEdit [4] further leveraged attribution to locate key visual layers and spatial regions relevant to prompts.

3 Method

3.1 Problem Formulation

Model editing aims to update or correct knowledge in pre-trained models without full retraining, and in the VLLM setting this objective extends to modifying multimodal behavior while preserving both textual and visual grounding. For a VLLM $f_\theta \in \mathcal{F}$, given an edit sample (x_i^e, x_t^e, y^e) such that $f_\theta(x_i^e, x_t^e) \neq y^e$, with x_i^e and x_t^e representing the image and textual modal inputs, and y^e being the pre-edit target output. Given a VLLM editor $\mathcal{E}_{\text{edit}}(\cdot) : (f_\theta, x_i^e, x_t^e, y^e) \rightarrow f'_\theta$, where $f'_\theta \in \mathcal{F}$ denotes the updated model, an effective multimodal knowledge editing method should satisfy the following three evaluation criteria.

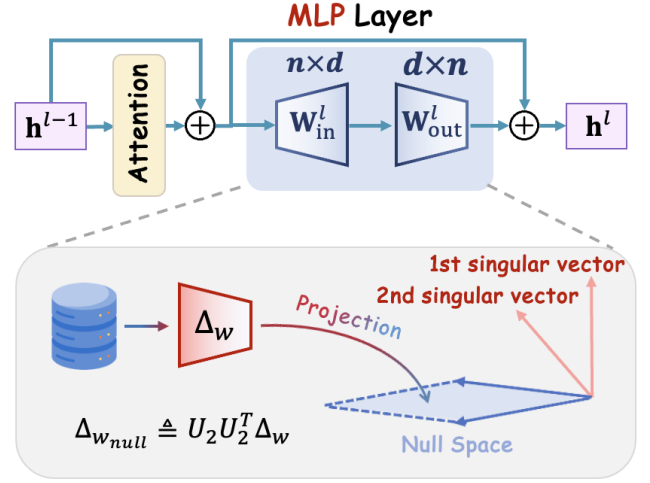


Figure 1: An illustration of our proposed method.

Reliability evaluates whether the post-edit model can produce accurate outputs for the edited instances, where $\mathcal{D}_e$ is the set of edit samples and $\mathbb{I}$ denotes the indicator function:

$$\mathbb{E}_{(x_i^e, x_t^e, y^e) \sim \mathcal{D}_{edit}} \mathbb{I} \{f_{\theta_e}(x_i^e, x_t^e) = y^e\} \quad (1)$$

Generality measures the model's ability to adapt to semantically or visually related variants of the edited sample in both multimodal and textual cases. $\mathcal{D}_{in}(x_i^e)$ and $\mathcal{D}_{in}(x_t^e)$ represent the textual and visual neighborhoods of the image x_i^e and prompt x_t^e :

$$\begin{aligned} &\mathbb{E}_{(x_i^e, x_t^e, y^e) \sim \mathcal{D}_e} \mathbb{E}_{x_t^{tn} \sim \mathcal{D}_{in}(x_t^e)} \mathbb{I} \{f_{\theta_e}(x_i^e, x_t^{tn}) = y^e\} \\ &\mathbb{E}_{(x_i^e, x_t^e, y^e) \sim \mathcal{D}_e} \mathbb{E}_{x_i^{in} \sim \mathcal{D}_{in}(x_i^e)} \mathbb{I} \{f_{\theta_e}(x_i^{in}, x_t^e) = y^e\} \end{aligned} \quad (2)$$

Locality requires the revised model to maintain consistent outputs with the original model on samples irrelevant to the edited instance, including both textual and visual locality. $\mathcal{D}_{tu}(x_i^e)$ and $\mathcal{D}_{iu}(x_t^e)$ denote the sets of unrelated text and image samples:

$$\begin{aligned} &\mathbb{E}_{(x_i^e, x_t^e, o^e) \sim \mathcal{D}_{edit}} \mathbb{E}_{(x_i^{tu}, o^{tu}) \sim \mathcal{D}_{tu}(x_i^e)} \mathbb{I} \{f_{\theta_e}(\emptyset, x_i^{tu}) = f_{\theta}(\emptyset, x_i^{tu})\} \\ &\mathbb{E}_{(x_i^e, x_t^e, o^e) \sim \mathcal{D}_{edit}} \mathbb{E}_{(x_i^{iu}, x_t^{iu}, o^{iu}) \sim \mathcal{D}_{iu}(x_t^e)} \mathbb{I} \{f_{\theta_e}(x_i^{iu}, x_t^{iu}) = f_{\theta}(x_i^{iu}, x_t^{iu})\} \end{aligned} \quad (3)$$

3.2 Covariance-Guided Directional Constraints

A central challenge in knowledge editing is to update specific facts without disrupting the model's behavior on unrelated inputs. This is especially difficult in VLLMs, where visual and textual modalities are tightly coupled. To reduce such interference, we constrain updates to directions unlikely to alter task-relevant representations. The key insight is that some parameter directions encode core knowledge, while others have little effect on output. By restricting updates to the latter, we introduce targeted knowledge while preserving behavior, ensuring lightweight yet reliable editing.

Given a layer input matrix $X \in \mathbb{R}^{n \times d}$, where n denotes the number of input tokens and d is the feature dimension, we compute its uncentered covariance matrix $C = X^T X$, which captures raw

Table 1: Performance comparison on E-VQA and E-IC datasets on BLIP2-OPT (2.7B) and LLaVA-V1.5 (7B)

Model	Method	E-VQA						E-IC					
		Rel.	T-Gen.	M-Gen.	T-Loc.	M-Loc.	Avg.	Rel.	T-Gen.	M-Gen.	T-Loc.	M-Loc.	Avg.
BLIP2-OPT	FT-V	23.00	15.41	19.08	97.87	86.97	48.47	40.14	38.66	34.65	98.94	88.39	60.16
	FT-L	23.63	15.86	19.94	96.75	88.41	48.92	40.00	38.15	35.41	98.02	87.46	59.81
	IKE	97.31	89.80	90.17	12.38	1.77	58.35	95.38	76.48	81.16	12.67	2.05	53.55
	SERAC	89.25	90.41	88.47	100.00	0.31	73.69	92.59	93.79	89.68	100.00	0.45	75.30
	MEND	90.59	89.86	90.46	94.52	63.74	85.83	64.23	35.88	34.21	90.99	55.07	56.08
	TP	66.64	59.39	55.04	97.50	83.77	72.47	47.03	46.72	42.92	91.65	79.16	61.50
	LTE	95.78	96.18	95.15	93.09	83.56	92.75	94.50	93.76	92.66	93.22	86.70	92.17
	VisEdit	96.66	96.74	97.39	100.00	90.04	96.17	95.82	95.02	93.66	100.00	90.63	95.03
	<i>LPEdit</i>	97.74	97.53	96.86	100.00	97.41	97.91	96.44	96.02	93.97	100.00	94.58	96.20
LLaVA-V1.5	FT-V	29.92	28.19	25.87	98.32	89.97	54.45	52.02	50.85	46.23	98.30	91.12	67.70
	FT-L	30.06	29.03	25.89	98.54	90.66	54.84	51.91	50.25	46.98	97.59	93.78	68.10
	IKE	89.88	89.15	89.17	59.32	50.56	75.62	92.49	86.66	78.46	74.88	64.11	79.32
	SERAC	80.78	79.86	78.87	100.00	56.17	79.14	41.23	39.98	40.99	100.00	7.29	45.90
	MEND	90.06	90.52	90.52	89.43	80.11	88.13	91.90	92.43	90.81	89.46	85.44	90.01
	TP	47.35	46.97	43.05	93.62	89.30	64.06	57.86	56.23	54.28	63.35	87.04	63.75
	LTE	92.99	92.12	91.57	81.73	79.70	87.62	92.04	90.77	89.78	84.38	87.21	88.84
	VisEdit	95.12	95.02	93.85	100.00	93.97	95.59	95.06	94.19	93.12	100.00	94.74	95.42
	<i>LPEdit</i>	95.15	94.98	93.92	100.00	98.39	96.49	95.23	94.50	93.43	100.00	97.41	96.11

correlations among input features without mean subtraction. Since not every covariance matrix possesses a strict null space in practice, we approximate it via singular value decomposition (SVD) [23].

To formalize, SVD factorizes the covariance matrix as

$$C = U\Sigma U^\top, \quad (4)$$

where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_d)$ with $\sigma_1 \geq \dots \geq \sigma_d \geq 0$. This admits the spectral expansion

$$C = \sum_{i=1}^d \sigma_i u_i u_i^\top, \quad (5)$$

where u_i are the singular vectors of C . Small singular values σ_i indicate weakly expressed representation directions, which form an approximate null subspace.

3.3 Low-Variance Projection for Editing

Building on the representation decomposition introduced above, we now describe how to constrain model updates using low-variance directions identified via singular value analysis. Specifically, we partition $U = [U_1, U_2]$ such that U_1 spans high-variance (informative) directions and U_2 spans near-zero singular directions. We therefore restrict model updates to the null subspace by projecting the gradient:

$$\Delta w_{\text{null}} = U_2 U_2^\top \Delta w, \quad (6)$$

ensuring that only low-energy directions are modified while the primary representational subspace remains intact.

We apply this projection to the MLP projection matrices in high-contribution layers, identified via consistently high gradient norms on the editing loss. MLPs are chosen over attention layers for their

localized role in intra-token transformations and factual encoding. The projection confines updates to the null space formed by low-singular-value directions, avoiding interference with the primary representational structure. By restricting edits to this low-impact subspace, the model incorporates new factual associations while keeping its responses to unrelated visual and textual inputs intact.

4 Experiments

4.1 Experiment Setting

Datasets: In line with Cheng et al. [6], we use E-VQA and E-IC as benchmark datasets for evaluating editing performance. E-VQA comprises 6,345 training and 2,093 testing examples from the VQA-v2 benchmark [1], where the model answers questions about images based on both visual and textual input. E-IC, derived from the COCO Caption dataset [5], targets caption correction and includes 2,849 training and 1,000 testing samples.

VLLM Backbones: To ensure a comprehensive evaluation on scale and architecture, we select two representative VLLMs for experimentation: BLIP2-OPT (2.7B) [12], LLaVA-V1.5 (7B) [13].

Evaluation metrics: We use Rel., T-Gen., M-Gen., T-Loc., M-Loc., and Avg. to denote Reliability, Textual Generality, Multimodal Generality, Textual Locality, Multimodal Locality, and the average score of these five metrics. All scores are reported in percentage format.

Baselines: Existing approaches primarily adapt LLM editors in the multimodal setting [6]. We include several representative baselines in our evaluation: FT-V (fine-tuning the visual encoder), TF-L (fine-tuning the final language model layer)[22], IKE [24], SERAC [18], MEND [17], TP [10], and LTE [11].

Table 2: Performance comparison on Qwen2.5-VL (7B)

Dataset	Method	Rel.	T-Gen.	M-Gen.	T-Loc.	M-Loc.	Avg.
E-VQA	TP	51.75	45.80	46.64	94.54	89.06	65.56
	LTE	93.86	92.83	92.25	89.19	85.60	90.75
	VisEdit	96.59	96.21	95.10	100.00	90.57	95.69
	<i>LPEdit</i>	96.81	97.63	97.74	100.00	99.27	98.29
E-IC	TP	57.49	58.27	55.93	67.72	78.16	63.52
	LTE	92.96	92.12	90.13	88.74	83.27	89.44
	VisEdit	95.22	94.48	92.98	100.00	90.26	94.59
	<i>LPEdit</i>	96.25	96.09	94.34	100.00	98.61	97.06

4.2 Analysis of Editing Performance

Table 1 presents the overall editing results. In the following, we provide a detailed experimental analysis from multiple perspectives. **In terms of methods**, *LPEdit* exhibits the strongest overall performance among all editing methods, particularly excelling in both locality metrics (T-Loc and M-Loc). VisEdit stands out as the most effective among existing baselines, leveraging precise visual-pathway manipulations across diverse evaluation dimensions. LTE achieves competitive results through fine-tuning, but shows limitations in maintaining stable performance on non-edited outputs.

In terms of datasets, editing on E-VQA generally yields higher performance, as the required corrections typically involve modifying short, discrete answers or a few key tokens in response to visual questions. These edits are often localized and less entangled with the broader input context, making it easier for the model to apply targeted updates. In contrast, E-IC requires full-sentence caption rewriting grounded in comprehensive image understanding, which poses greater challenges for maintaining locality.

In terms of models, most editors demonstrate greater stability on BLIP2-OPT, likely due to clearer modular separation between visual and language pathways, allowing local interventions to remain more contained. In contrast, LLaVA-V1.5 integrates visual features more deeply into the language decoder, making localized editing more difficult and leading to increased performance variation among editors. This highlights the importance of architectural compatibility in multimodal editing. Qwen2.5-VL exhibits consistently higher editing robustness relative to LLaVA, suggesting that its modular interaction between visual encoders and the language backbone facilitates localized parameter adjustments.

In terms of evaluation metrics, *LPEdit* achieves the highest performance on both textual and visual locality metrics, indicating strong consistency on unrelated samples across modalities. FT-V and VisEdit, which apply edits to the visual modality, naturally preserve the language generation pathway and maintain linguistic locality and overall output quality. SERAC leverages a classifier to identify purely textual inputs, which helps maintain locality on the language side but oversees visual samples. Methods with lower overall performance often exhibit high locality at the expense of reliability and generality, suggesting that these objectives are difficult to satisfy simultaneously. In contrast, *LPEdit* demonstrates a more balanced performance: it significantly enhances locality while maintaining competitive reliability and generality.

5 Conclusions

We present *LPEdit*, a locality-preserving method using null-space projection for multimodal knowledge editing. By constraining parameter updates to directions that minimally affect unrelated knowledge, our method achieves accurate edits while preserving model behavior on non-target information. Experimental results across multiple models and tasks demonstrate *LPEdit*'s effectiveness and generalization ability, showcasing its potential to make reliable and general knowledge editing in multimodal systems.

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